

DISTRICT-LEVEL MAPPING AND HAZARD ASSESSMENT OF EXTREME RAINFALL IN MAHARASHTRA USING EXCEEDANCE PROBABILITY ANALYSIS

A Group Assignment Report

(CM801 – INTRODUCTION TO RISK ANALYSIS)

submitted by

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Introduction:

The hydrological cycle is the continuous process of recycling water through the atmosphere, land, and oceans, involving evaporation, condensation, transpiration, precipitation, runoff, and infiltration. Precipitation is the most important component of the water cycle, as it is a major source of water on the planet, second only to glacier melt. Even though precipitation is important for sustaining life for all living beings on the planet, extreme amounts can cause flooding and storms that damage crops, property, the economy, and lives. Extreme rainfall events are increasingly frequent due to climate change and atmospheric dynamics. Understanding the spatial distribution and intensity of extreme rainfall is crucial for disaster preparedness and risk mitigation. Remote sensing and gridded rainfall datasets provide an efficient way to analyse such events over large areas and extended time periods.

This study focuses on identifying rainfall hazard zones across Maharashtra by analysing temporal rainfall data and deriving statistical indicators such as maximum rainfall. The results aim to support decision-making in disaster management and regional planning.

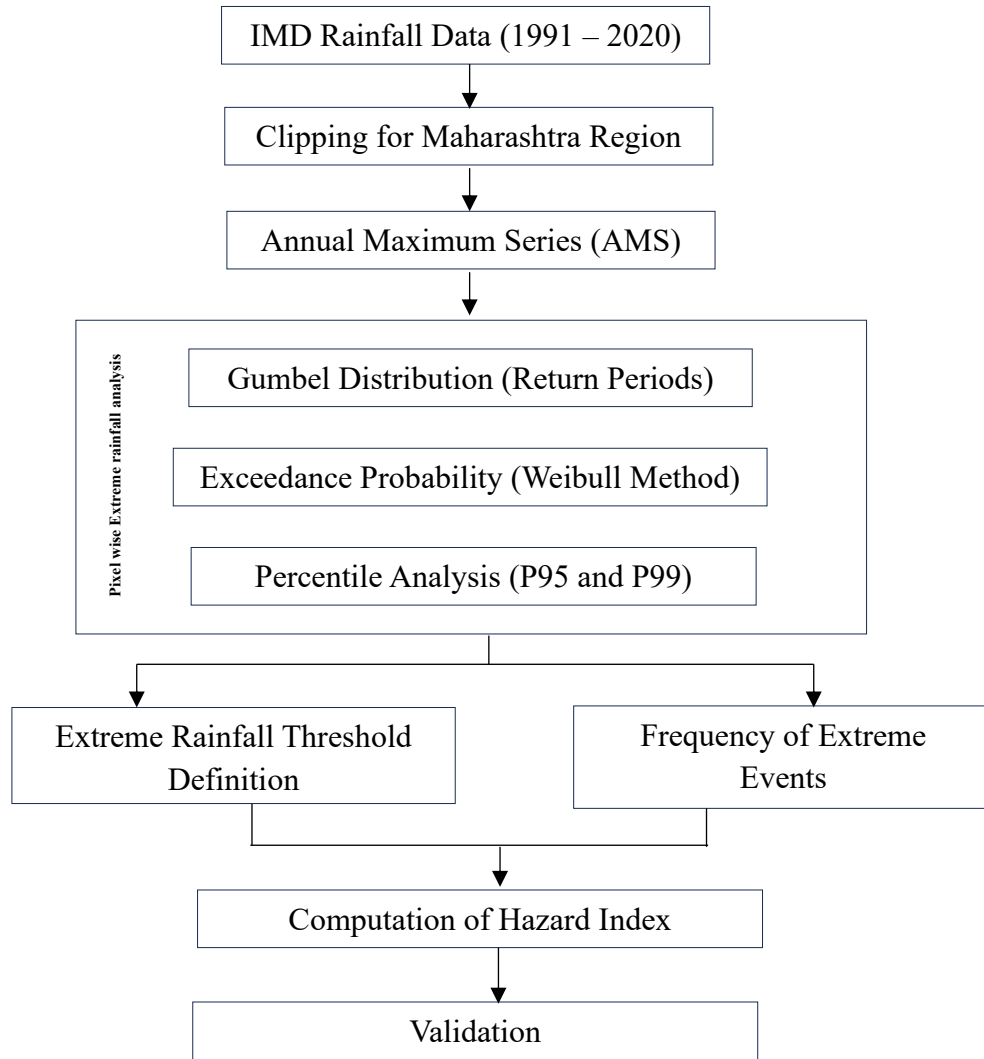
Data and Methods:

Precipitation data for 30 years, from 1991 to 2020, has been downloaded for the Indian subcontinent from the Indian Meteorological Department (IMD), and it is clipped for Maharashtra state using the Maharashtra state shapefile downloaded from the Survey of India web portal. The horizontal and vertical resolution of the IMD gridded rainfall dataset is 27.75 km.

For each pixel in the Maharashtra state, the maximum rainfall for each year has been computed. And then the average of the maximum rainfall across all years has been computed; this is the Annual Maximum Series, used to obtain information on extreme events. Pixel-wise extreme rainfall analysis was performed by fitting a Gumbel (Type I Extreme Value) distribution to the 30-year Annual Maximum Series at each grid cell. Rainfall estimates were derived for return periods of 2, 5, 10, 25, and 50 years. To translate pixel-level extreme rainfall estimates into administratively meaningful units, zonal statistics were computed using Maharashtra's district boundaries. Similarly, the AMS data has been translated into district-level zonal statistics.

For our study, we have defined the thresholds for the 95th and 99th percentiles of extreme rainfall. Pixel-wise empirical extreme value analysis using Weibull plotting position has been performed on the data, which has the maximum rainfall for all 30 years. In this step, the annual rainfall has been ranked and assigned a probability (exceedance probability) and return period. Weibull plotting positions provide a nonparametric basis for comparison with Gumbel-derived extreme-value estimates.

Empirical return periods derived from Weibull plotting positions were compared with Gumbel-based estimates to assess the suitability of the distribution. Empirical return period rainfall maps using the Weibull method have been created.



Hazard is the function of frequency and intensity. A composite hazard index was developed by integrating extreme rainfall intensity (99th percentile) and frequency (exceedance counts), with a higher weight assigned to intensity. The resulting spatial patterns highlight regions prone to both intense and frequent extreme rainfall events. District-level hazard was derived by aggregating pixel-wise hazard index using zonal statistics. The results highlight spatial variability in rainfall extremes, identifying districts with elevated risk due to both high intensity and frequency of extreme rainfall event.

Results and Discussion:

Annual Maximum Rainfall Analysis:

From the mean rainfall map given in Figure 1, regions located along the Western Ghats and Konkan coast record significantly higher extreme rainfall compared to the interior Deccan plateau.

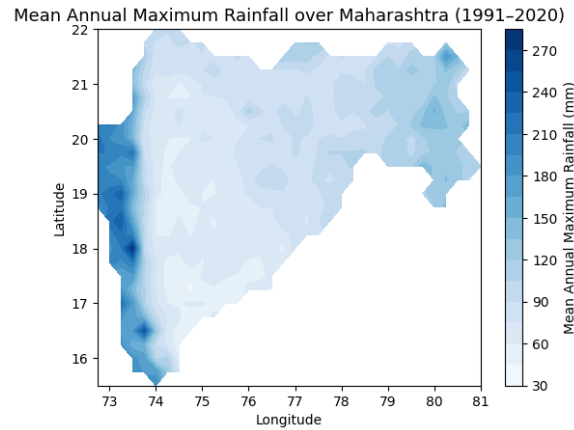


Figure 1 – Mean Annual Maximum Rainfall Over Maharashtra

From Gumbel Return levels, district wise statistics (Figure 2) has been computed which indicates that the coastal districts are having significantly higher rainfall in all the return periods.

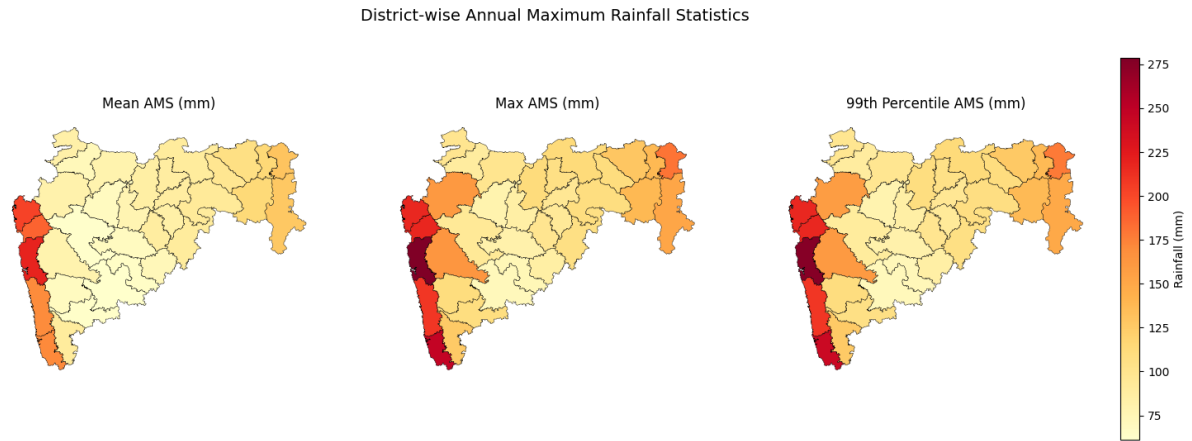


Figure 2 – District wise Annual Maximum Rainfall Statistics Map

The district level AMS analysis indicates that the mean of the rainfall is uniform across the state however spatial variability is observed in the 99 percentiles. This indicates that extreme events, rather than average rainfall, drive spatial differences in rainfall behavior. The dominance of high percentile values in coastal regions further justifies the use of extreme-based metrics (e.g., P99) for hazard assessment.

Gumbel Return Level Analysis:

District-level aggregation of return levels reveals a consistent pattern: **coastal districts such as Ratnagiri, Sindhudurg, Raigad, and Mumbai** dominate across all return periods (Figure 3). These districts exhibit the highest rainfall magnitudes, especially for higher return periods, indicating their vulnerability to extreme events. Even at lower return periods, the coastal districts show an elevated rainfall.

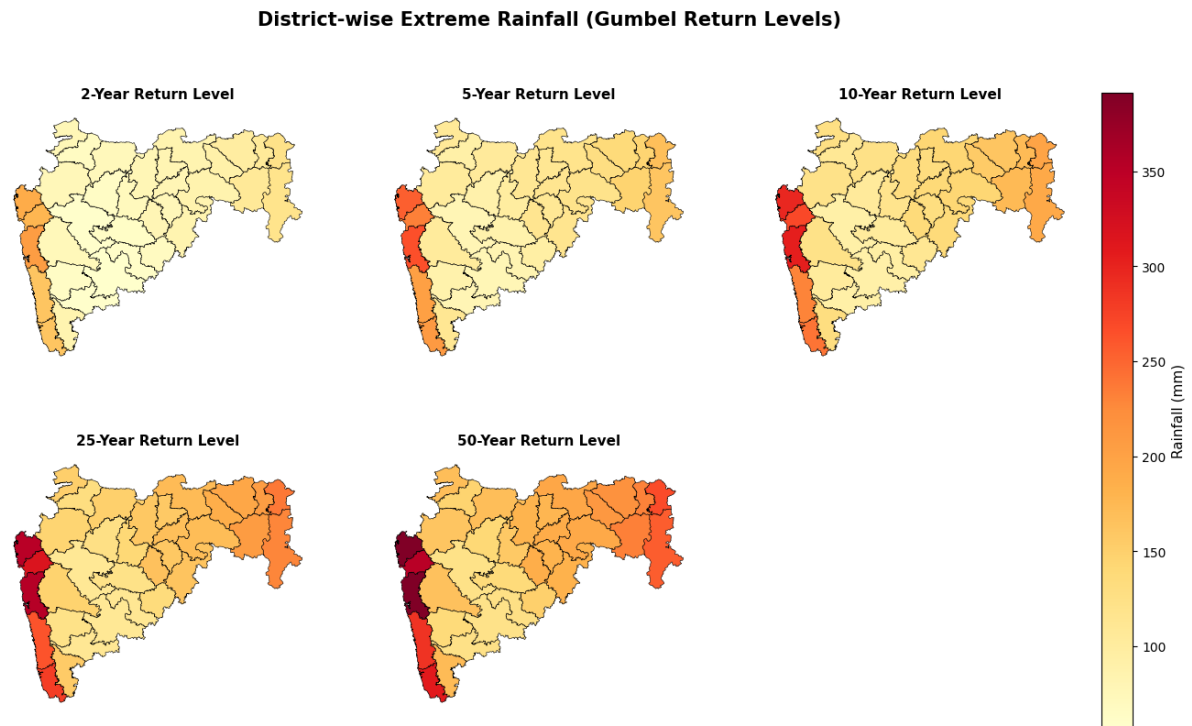


Figure 3 – District-wise Extreme Rainfall Analysis

The increase in rainfall with return period is non-linear, with a sharper rise observed in regions with already high rainfall. This indicates that areas receiving high rainfall under normal conditions are also more susceptible to extreme intensification, particularly along the Western Ghats.

Exceedance Probability Analysis:

A comparison between P95 and P99 thresholds indicates that P95 values are tightly clustered, while 9PP values exhibit a wider spread and higher magnitudes (Figure 4). This suggests that rare extreme events (P99) are not only more intense but also more spatially variable. Therefore, using higher percentiles provides a more realistic representation of extreme rainfall behaviour.

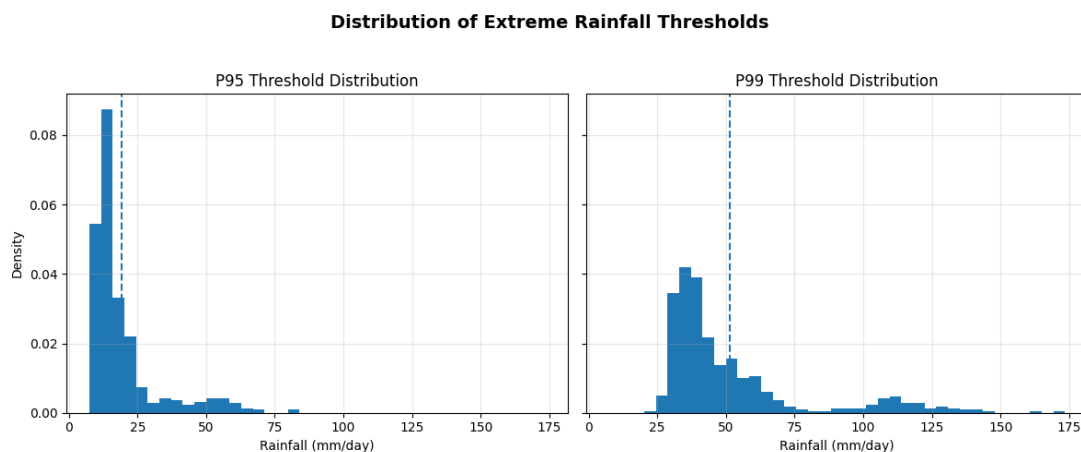


Figure 4 – Comparison of P99 and P95 Extreme Rainfall Thresholds

Hence, P99 is used to obtain the exceedance probability values. The comparison between the Weibull (empirical) and Gumbel (theoretical) approaches shows that, at higher return periods, Gumbel tends to slightly smooth extreme values, whereas Weibull reflects isolated high-intensity events. Despite these differences, both methods exhibit consistent spatial patterns, reinforcing the reliability of the results.

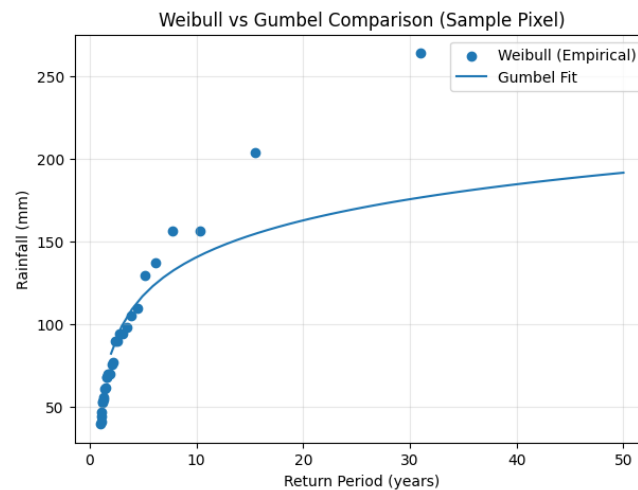


Figure 5 – Weibull Plotting VS Gumbel Distribution

Weibull-based return level maps show similar spatial trends as Gumbel maps, with high values concentrated along the western coast. However, the Weibull maps display greater pixel-level variability, indicating sensitivity to local rainfall extremes (Figure 5 and 6).

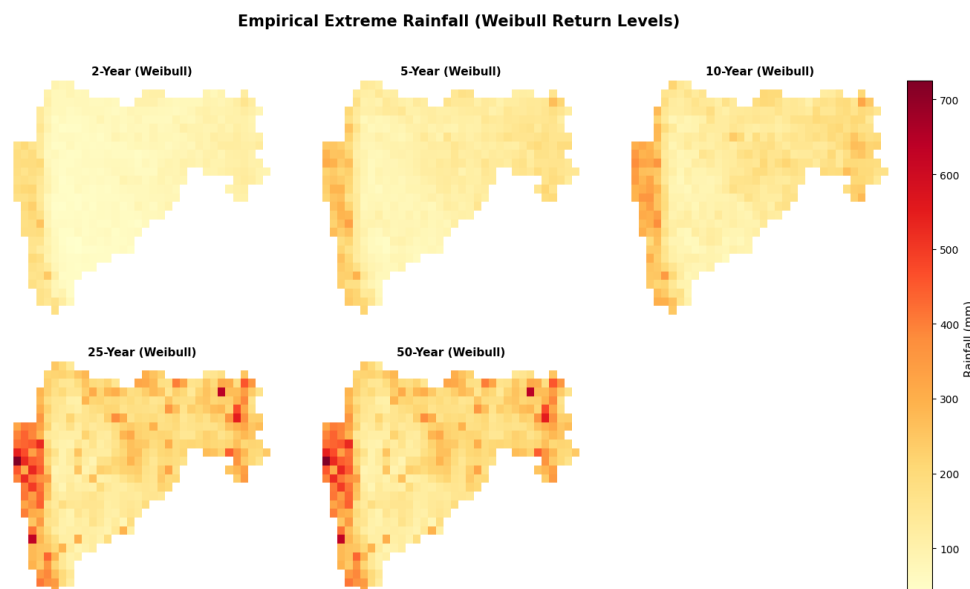


Figure 6 – Empirical Extreme Rainfall maps for different return periods using Weibull Plotting

Hazard Mapping:

The extreme rainfall intensity map, computed as a weighted combination of $0.8 \times P99$ and $0.2 \times P95$, highlights the dominance of high-intensity rainfall along the western coastal belt and Western Ghats region. These areas exhibit the highest rainfall magnitudes due to orographic effects, while the interior regions show comparatively lower intensity. The frequency map, based on the number of days exceeding the P99.5 threshold, indicates that extreme rainfall events occur more frequently in the same western regions, with moderate frequency extending toward central Maharashtra. This suggests that not only are rainfall events intense in these areas, but they also occur more often. The rainfall hazard map, derived using a weighted combination of $0.6 \times \text{intensity}$ and $0.4 \times \text{frequency}$, shows a clear spatial pattern (Figure 7).

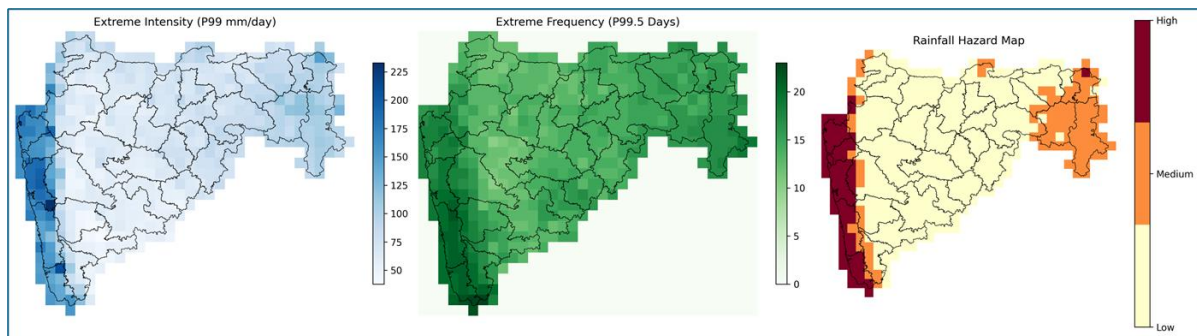


Figure 7 – Intensity maps and pixel-wise hazard map

The district-level hazard classification further confirms the spatial pattern observed at the pixel level.

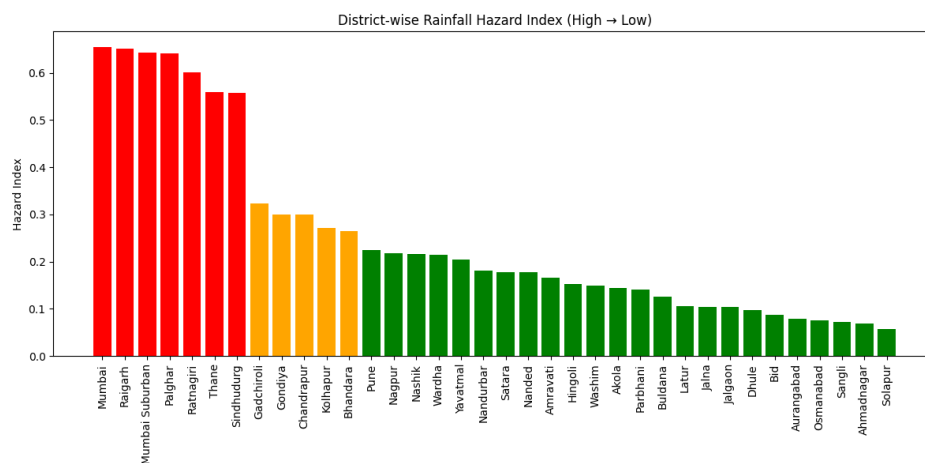


Figure 8 - Ranking of Districts on the basis of Hazard (High to Low)

The highest hazard zones are concentrated along the western coastal districts, including regions influenced by the Western Ghats (Figure 8 and 9). The central parts of Maharashtra fall under moderate hazard, while the eastern regions generally exhibit lower hazard levels. The dominance of the western region in the hazard map indicates that rainfall intensity plays a slightly stronger role than frequency in determining hazard, as reflected by the higher weight (0.6). However, including frequency ensures that areas with repeated extreme events are also captured. Overall, the results demonstrate that regions experiencing both high-intensity and

frequent extreme rainfall are identified as the most vulnerable, confirming the effectiveness of the combined hazard index approach.

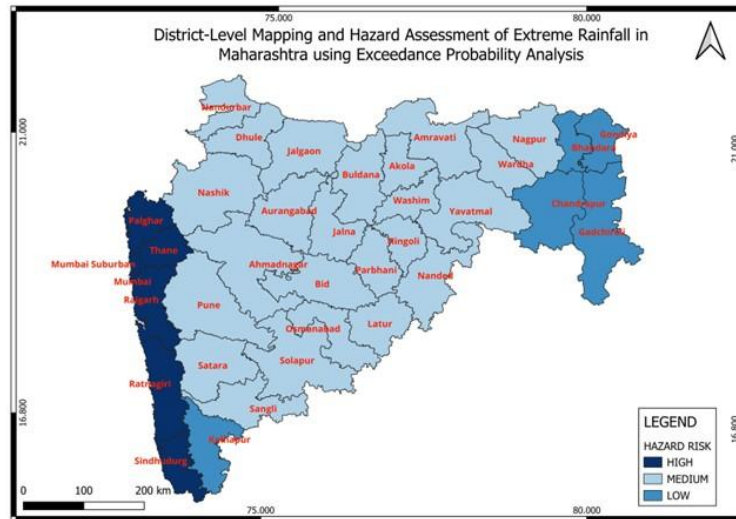


Figure 9 – Rainfall Hazard Map for Maharashtra

Validation:

Precipitation data from the Indian Meteorological Department for 2021 have been downloaded for the Maharashtra region, and the extreme rainfall values (99th percentile) are used for each pixel. They are correlated with the hazard index values for each pixel, yielding a strong correlation of 0.73.

The sensitivity analysis was conducted by varying the weights assigned to rainfall intensity and frequency in the hazard index (0.5–0.5, 0.6–0.4, and 0.7–0.3). The resulting hazard maps exhibit extremely high correlation values ($r = 0.995$ – 0.999), indicating that the spatial distribution of hazard remains nearly unchanged across different weighting schemes.

The difference map further shows that most variations are minimal (generally < 0.02), with slightly higher differences observed only in the Western Ghats region. However, these variations do not significantly alter the overall hazard pattern or classification.

Overall, the analysis demonstrates that the hazard model is highly robust and insensitive to moderate changes in the weighting scheme.

CONCLUSION:

Extreme rainfall hazard is spatially concentrated along the western coast. Rainfall intensity plays a dominant role in hazard formation. Extreme rainfall increases non-linearly with return period. Both empirical and theoretical approaches show consistent spatial patterns. Hazard distribution is highly uneven, with coastal districts being the most vulnerable

REFERENCES:

- 1) Nathan, R., Jordan, P., Scoria, M., Lang, S., Kuczera, G., Schaefer, M., & Weinmann, E. (2016). Estimating the exceedance probability of extreme rainfalls up to the probable maximum precipitation. *Journal of hydrology*, 543, 706-720.